PV101 Day 1: First Day Orientation

CTS Cohort 6 | Monday, 2026-07-06 | Full Instructor Packet

Instructor Snapshot

Day 1 Theme

Solar work starts with orientation, shared language, safe habits, and verification. The goal is not to overload students with math. The goal is to make the room safe enough to learn and practical enough to matter.

Learning Targets

- Explain what PV means and why solar work starts with observation, labels, and safe habits.
- Name the basic parts of a small PV system: solar module, charge controller, battery, fuse, load, and inverter.
- Describe the class rule: measure and verify before trusting a label.
- Practice group roles, cleanup expectations, and first-day classroom agreements.
- Connect Day 1 to the Cohort 6 RV solar capstone.

Materials

- Whiteboard and markers
- Printed student packet
- Printed Day 1 quiz and exit ticket
- Pictures or physical examples: solar module, charge controller, battery, fuse holder, inverter, simple loads
- Cohort 6 classroom default battery note: 12.8V LiFePO4 battery is the available training battery for later meter practice
- Tape or sticky notes for group system maps

Full Day Flow

Time	Block	Instructor Notes
10:00-10:15	Arrival, names, room reset	Students settle in, instructor checks energy level, materials out.
10:15-10:35	Welcome to PV101	What PV means, why this class matters, and how the RV capstone fits.
10:35-11:00	Class agreements	Safety, respect, asking for help, no public shaming, cleanup is part of the job.
11:00-11:20	What is a solar system?	Module -> charge controller -> battery -> fuse/load -> inverter when needed.
11:20-11:30	Break	Reset, water, bathroom.
11:30-12:05	Solar scavenger hunt	Groups identify parts from pictures or physical examples.
12:05-12:30	RV capstone preview	What in an RV needs power? Students

		list loads and questions.
12:30-1:00	Lunch	No work during lunch unless students choose to keep discussing.
1:00-1:35	Labels vs verification	Why labels help but do not replace testing and tracing.
1:35-2:10	Group build: system map	Students draw the energy path for a simple RV solar system.
2:10-2:30	Quiz	Short low-pressure check for teaching quality and vocabulary.
2:30-2:50	Exit ticket	What makes sense, what is confusing, what students want to learn.
2:50-3:00	Cleanup and preview Day 2	Meter readiness and electricity basics tomorrow.

Opening Script

- Welcome students by name when possible. Keep the first few minutes low pressure.
- Say: "This class is practical. We are going to learn how solar power actually moves through a small system, especially the kind you could see in an RV, trailer, shop, or off-grid setup."
- Say: "The most important rule in this room is simple: we measure and verify. We do not guess and we do not shame people for not knowing yet."
- Say: "A quiz here tells me how well I taught. It is not a trap."

Class Agreements

Agreement	Plain Meaning
Ask before touching unknown equipment	Curiosity is good. Random touching is not a curriculum.
No public shaming	Wrong answers are information, not entertainment.
Tools return home	A tool that cannot be found is not a tool. It is a future problem.
Clean as you go	The job is not done until the workspace is ready for the next person.
Use names and roles	Groups work better when everyone knows who is doing what.
Pause when unsure	Stopping is a skill. Guessing is how systems become mysteries.

Core Board Diagram

Draw this on the board

Sun -> Solar Module -> Charge Controller -> Battery -> Fuse Panel -> DC Loads

Battery -> Inverter -> AC Loads

Teach the Diagram

- **Sun:** The energy source. No sun, less production.

- **Solar module:** Turns light into DC electrical energy.
- **Charge controller:** Protects the battery by controlling charging.
- **Battery:** Stores energy. For Cohort 6, use 12.8V LiFePO4 as the future worksheet default.
- **Fuse panel:** Distributes power and protects circuits.
- **Loads:** Anything that uses power: lights, fan, water pump, USB ports, appliances.
- **Inverter:** Optional part that changes DC battery power to AC power.

Group Activity: RV Load Brainstorm

- Put students in two groups of three if all six are present.
- Prompt: "What in an RV might need electricity?"
- Require each group to list at least eight possible loads.
- Then have them mark each as likely DC, likely AC, or unknown.
- Do not worry about perfect answers yet. The point is to make the system feel real.

Possible RV Load	Likely Power Type	Why it matters
Lights	DC	Common low-voltage load.
Water pump	DC	Often hidden behind a fuse or switch.
Roof fan	DC	Comfort load with real runtime impact.
USB charger	DC	Small but common.
Mini fridge	DC or AC	Depends on model.
Microwave	AC	High power draw.
Laptop charger	AC or DC adapter	Depends on charger.
Inverter outlets	AC	Powered from battery through inverter.

Labels vs Verification Mini-Lesson

- Labels are helpful clues, not proof.
- Old systems get modified. People leave surprises behind. Future troubleshooting depends on clear documentation.
- A good solar worker asks: What do I see? What does the label say? What can I verify? What still needs tracing?
- Connect this to the future RV capstone: the goal is not just to make it work today. The goal is to make it serviceable later.

Day 1 Formative Questions

- What does PV mean?
- What part stores energy?
- What part controls charging?
- What is a load?
- Why do we verify instead of guessing?
- What is one RV load you already understand?

Day 2 Preview

Tomorrow

Day 2 moves into electricity basics and multimeter readiness. Future worksheet versions should use the 12.8V LiFePO4 battery as the default available battery and should not assume AA, AAA, or 9V batteries are on hand.